Dr. Jinxin Li

Physicist & Software Engineer

Scientific Computing · C++ / Python · AI Systems

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Professional Profile

Experimental particle physicist and software engineer with nearly seven years of scientific software and research experience using C++ and Python. Combines quantitative reasoning with simulation, reconstruction, statistical analysis, algorithm design and framework maintenance in Linux/HPC environments. Extends this foundation into AI systems through PANDA Agent, a research assistant connecting code-aware retrieval, backend workflows and systematic evaluation, with ownership of architecture, requirements and technical acceptance.

Technical Skills

Programming & engineering: C++, Python, Bash/Shell; object-oriented design, Git, Linux, Docker, debugging, integration, automated testing and regression validation.

Scientific computing & HPC: ROOT, PandaRoot, FairRoot, NumPy, Pandas, Scikit-learn, SLURM, Singularity; Monte Carlo simulation and statistical analysis.

AI systems: RAG, hybrid dense/BM25 retrieval, reranking, evidence grounding and evaluation.

Backend project exposure: LangGraph, FastAPI / REST APIs, PostgreSQL, Qdrant, SQLAlchemy.

Languages: Chinese (Native), English (Proficient), German (CEFR A2).

Professional Experience

Doctoral Researcher / Scientific Software Developer

Ruhr University Bochum - Institute of Experimental Physics I, PANDA Collaboration
Sep. 2019 - Jul. 2026 · Bochum, Germany

- Developed C++ and Python software for particle tracking, detector simulation, reconstruction and data analysis in Linux/HPC environments.
- Served as Software Owner for the PANDA Luminosity Detector Group, maintaining, debugging and integrating scientific software.
- Contributed functionality, reconstruction extensions and bug fixes to PandaRoot through collaborative, cross-component development.
- Established automated validation and technical documentation; delivered training and mentored undergraduate thesis work.

Selected Engineering Projects

PANDA Agent - AI Research Assistant & Code Intelligence

Apr. 2026 - Present · Independent project · github.com/JinxinLee/PANDA_Agent

- Architected a local research prototype with version-aware code and literature ingestion, cross-repository hybrid dense/BM25 retrieval, reranking and citation verification.
- Designed and integrated Python backend workflows with LangGraph, FastAPI, PostgreSQL and Qdrant through AI-assisted development, retaining requirements, review and technical acceptance ownership.
- Designed versioned, human-reviewed benchmarks with separate development and validation datasets and an externally managed sealed-holdout protocol.
- Established reproducible retrieval/QA evaluation, controlled ablations and regression checks, with structured execution traces and diagnostics for failure analysis.

Selected Engineering Projects (continued)

LuminosityFit - C++ Framework Development

Mar. 2022 - Jul. 2024 · Doctoral research

- Primary maintainer and technical owner of an existing C++ fitting framework; extended numerical workflows for distributed interaction regions and beam parameters, with debugging and PandaRoot integration.
- Integrated reconstructed target distributions into an end-to-end reconstruction and fitting chain, achieving **(0.05 ± 0.07)% relative luminosity deviation** from simulated ground truth under the studied conditions.

Particle Reconstruction & Iterative Correction

Jul. 2024 - Apr. 2026 · Doctoral research

- Developed a POCA-based vertex workflow integrating existing tracking and GenFit/Kalman components; achieved **74.4% event reconstruction efficiency** versus a 74.2% true-vertex reference in a controlled displaced-interaction simulation.
- Developed simulation-based efficiency correction and iterative refinement, reducing global target-profile discrepancy from **12.27% at iteration 1 to 1.30% at iteration 3** in a simulation stress test.

Detector Simulation & Performance Evaluation

Sep. 2019 - Mar. 2022 · Doctoral research

- Implemented an alternative HV-MAPS detector geometry in PandaRoot and benchmarked tracking, vertex resolution and event reconstruction against the baseline detector design.
- The detector-design simulation showed **2.8× higher D-meson reconstruction efficiency** (3.33% to 9.26%) and **44-52% smaller primary-vertex resolution widths** across the three coordinate axes.

Education

Dr. rer. nat. (Ph.D.) in Physics

Ruhr University Bochum, Germany · 2026

Dissertation: *Luminosity Determination with Restgas Background from the Target for the PANDA Experiment*

B.Sc. in Physics

Nankai University, Tianjin, China · Sep. 2015 - Jun. 2019

Earlier Research Experience

Institute of High Energy Physics, Chinese Academy of Sciences - BESIII

Research Assistant · Jun. 2018 - Jul. 2019

Charm/D-meson analysis, Monte Carlo simulation and ROOT-based statistical analysis.

Nankai University - BESIII

Undergraduate Research Assistant · Sep. 2016 - Jul. 2019

Charmonium analysis, simulation and data processing.

AMS-02 Project (CERN / MIT collaboration)

Summer Research Assistant · Jun. 2017 - Sep. 2017

Detector simulation and tracker-resolution validation.